Exact negative-Einstein spectral degeneration on non-virtually-Kähler manifolds

23 September 2026. This note proves a global escape alternative for the allowed branch. It does not assert that small Weyl splitting always vanishes. The main argument combines the previously proved signed harmonic-Weyl estimate with classical smooth Einstein compactness and an actual smooth-topological obstruction. Its scope is all exact outputs satisfying the stated geometric bounds, not a neighbourhood of a selected approximate construction.

Strengthening for the actual aspherical family. The subsequent aspherical no-bubble theorem removes the opposite-Weyl bound from the self-gluing application below. In that family, vanishing splitting forces diameter escape alone. Theorem 1 and the general degeneration discussion remain useful for other non-virtually-Kähler types where asphericity is not available.

1. Hypotheses and the new conclusion

Let M be a closed connected oriented smooth four-manifold for which no connected cover of degree at most two admits a Kähler structure compatible with the lifted orientation. Suppose there is $A > 0$ such that each metric under consideration has an actual polarized monopole class a_g with $(a_g^+)^2 \geqslant A$. The class may depend on the metric. The scalar monopole estimate, not an assumption about a formal cohomology vector, is required.

Consider exact metrics $\mathrm{Ric}_g = -g$. Order the eigenvalues of W_g^+ and set

$$\lambda_1 < \lambda_2 \leqslant \lambda_3, \qquad t = -\frac{3}{4}\lambda_1, \quad b = \lambda_3 - \lambda_2, \quad z = \frac{b}{4t}.$$

Fix $m > 0$ and $0 \leqslant \theta < 1/\sqrt{3}$, and impose

$$t \geqslant m, \qquad z \leqslant \theta. \tag{1}$$

Define the normalized splitting energy

$$\mathscr{S}(g) = \fint_M b^2.$$

Theorem 1 (global bounded-geometry splitting gap). For every $D, K > 0$ there is $\varepsilon = \varepsilon(A, m, \theta, D, K) > 0$ such that every metric above satisfying

$$\mathrm{diam}(M, g) \leqslant D, \qquad \|W_g^-\|_\infty \leqslant K \tag{2}$$

satisfies

$$\mathscr{S}(g) \geqslant \varepsilon. \tag{3}$$

The constant can be chosen uniformly over all smooth types satisfying the same hypotheses and the same positive budget A. No independent volume, injectivity-radius, full-curvature or Sobolev hypothesis is needed.

Corollary 2 (escape alternative). On a fixed such M, if exact negative-Einstein metrics satisfy (1) and $\mathscr{S}(g_j) \to 0$, then

$$\mathrm{diam}(M, g_j) + \|W_{g_j}^-\|_\infty \longrightarrow \infty. \tag{4}$$

More precisely, every fixed bounded rectangle in these two quantities contains only metrics with a positive lower splitting energy. It is not claimed that each summand in (4) tends to infinity separately.

This conclusion changes a global decision at the canonical shear ratio $T/A = 1$: the earlier monopole splitting bound proportional to $(1 - T/A)^2$ vanishes there, while (3) is strictly positive under (2). It does not exclude arbitrary outputs escaping (2).

2. Derivation of the geometric compactness, with no redundant inputs

Since $s = -4$, the actual scalar monopole inequality gives

$$16\,\mathrm{Vol}(g) = \int_M s^2 \geqslant 32\pi^2 (a_g^+)^2 \geqslant 32\pi^2 A, \qquad \mathrm{Vol}(g) \geqslant 2\pi^2 A. \tag{5}$$

The maximum-principle result in the Einstein spectral application gives

$$t \leqslant M_\theta = \frac{1}{2(1 - 3\theta^2)}. \tag{6}$$

Moreover,

$$|W^+|^2 = \frac{8}{3} t^2 + \frac{1}{2} b^2 \leqslant \left(\frac{8}{3} + 8\theta^2 \right) M_\theta^2. \tag{7}$$

Consequently the bound on W^-, together with the Einstein equation, gives a full sectional-curvature bound depending only on θ, K.

With $\mathrm{Ric} = -g$, Bishop comparison and the diameter bound also give an upper volume bound depending only on D. In addition, relative Bishop comparison and (5) give a uniform lower bound for the volume of every ball of any fixed positive radius at most D. The sectional bound and these bounds imply the standard uniform injectivity-radius estimate. Thus every sequence satisfying (1)–(2) has, after a subsequence and pull-back by orientation-preserving diffeomorphisms, a smooth Einstein limit on a closed smooth manifold. Here smoothness follows by elliptic bootstrapping of the exact Einstein equation after Cheeger–Gromov compactness. This is classical compactness, not a new regularity theorem.

3. Proof of the splitting gap

If (3) failed, take a sequence with $\mathscr{S}(g_j) \to 0$ and use the preceding smooth compactness. The ordered Weyl eigenvalues converge uniformly. In the limit, $b_\infty = 0$ everywhere and $t_\infty \geqslant m$. Thus the lowest Weyl eigenline remains smooth and separated from its two-dimensional complement.

One can already finish with the exact Derdziński theorem. Equivalently, the proved signed harmonic-Weyl estimate gives, for the normalized lowest eigenform on its orientation cover,

$$⨍ \left(|dt_j|^2 + (t_j - \tfrac{1}{2})^2 + |\nabla \omega_j|^2 \right) \leqslant C_{m,\theta} \mathscr{S}(g_j). \tag{8}$$

The line bundles converge under the diffeomorphisms: the uniform spectral gap makes their spectral projections smoothly convergent, and sufficiently close real line bundles are isomorphic. Their orientation covers are therefore identified with the limit orientation cover. This is a cover of degree at most two, not an assumed global choice of eigenform on M.

Passing to the smooth limit in (8) gives $t_\infty = 1/2$ and a parallel self-dual form $|\omega_\infty|^2 = 2$. The corresponding orthogonal almost complex structure is parallel, hence integrable and Kähler. The

lifted metric is negative Kähler–Einstein. Since smooth convergence preserves the diffeomorphism type, this contradicts the topological hypothesis for every sufficiently late manifold of the sequence. This also proves the claimed uniformity across smooth types. Corollary 2 is its contrapositive.

4. Actual application and removal of the lower amplitude assumption

For the self-gluings

$$M = M(q, g; u, f), \qquad (\phi, x) \longmapsto (-\phi + u(x), f(x)), \quad [u] \neq 0, \quad q \geqslant 1, \quad g \geqslant 2, \quad f \in \mathrm{Diff}^+(\Sigma_g),$$

the inherited virtual Kähler theorem excludes all finite Kähler covers. The canonical-class calculation supplies an actual canonical monopole class a with

$$a^2 = 2\chi(M) > 0, \qquad (a_g^+)^2 \geqslant a^2.$$

Thus Theorem 1 applies with $A = 2\chi(M)$, to every negative Einstein metric in its stated spectral class, independently of the original gluing coordinates or of its position in the Einstein moduli space.

For these objects $\tau = 0$ and $T = A$. Hence the previously proved topological splitting estimate involving $(1 - T/A)^2$ is zero. The present positive bound instead uses the obstruction to a degree-two Kähler cover. It is therefore not an algebraic restatement of that earlier estimate. No assertion is made that this compactness corollary is new in the published literature.

The known almost-Einstein sequences on these manifolds do not contradict the theorem: they are not exact Einstein and their long thin geometry leaves the compact regime. The result still permits a non-Kähler exact Einstein metric of appreciable Weyl splitting.

In fact this actual family allows removal of the lower amplitude bound. This strengthening uses both its signature and its fundamental group, not an assumption that the selected eigenline cannot disappear.

Theorem 3 (shear-family gap without lower Weyl amplitude). Fix $A_0 > 0$, $D, K > 0$, and $0 \leqslant \theta < 1/\sqrt{3}$. For every member of the shear family with $2\chi(M) \geqslant A_0$, every exact $\mathrm{Ric}_g = -g$ metric satisfying

$$b \leqslant 4\theta t, \qquad \mathrm{diam}(g) \leqslant D, \qquad \|W^-\|_\infty \leqslant K$$

has $\mathscr{S}(g) \geqslant \varepsilon(A_0, \theta, D, K) > 0$. Here ordered eigenvalues define t, b everywhere, and $W^+ = 0$ is allowed. There is no prescribed $m > 0$. In particular, any exact sequence in this relative pinching class with vanishing splitting energy must satisfy the diameter/opposite-Weyl escape conclusion (4). The later Theorem 3 strengthens this to $\mathrm{diam}(g_j) \to \infty$, with a uniform splitting gap depending only on D, θ, and without assuming K.

Proof. At any positive maximum of t, the lowest eigenvalue is simple because $\lambda_2 - \lambda_1 = 2t - b/2 \geqslant 2(1 - \theta)t > 0$. The previous maximum-principle calculation therefore proves (6) even when W^+ has zeros. If $t \equiv 0$, (6) is immediate. Thus all the compactness conclusions of Section 2 remain available, uniformly using A_0 in (5).

If a gap failed, the smooth limit would be a compact negative Einstein metric with $\lambda_2 = \lambda_3$ everywhere. Derdziński's Proposition 5(iv) and Theorem 2 give two relevant alternatives: $W^+ \equiv 0$, or a Kähler–Einstein metric on a cover of degree at most two. The possible non-Kähler conformally Kähler case of that classical classification has a nonzero Killing field; a compact negative Einstein

metric has none by the integrated Bochner formula. Thus it does not provide another case here. The KE alternative contradicts the virtual Kähler obstruction.

In the ASD alternative, the inherited signature $\tau(M) = 0$ and the signature formula give $\int |W^-|^2 = 0$. Both Weyl halves vanish, so the limit has constant sectional curvature $-1/3$. But the injected vertex group $\pi_1(F_{q,2}) \times \pi_1(\Sigma_g)$ contains $\mathbb{Z}^2$. This contradicts Preissmann's theorem for closed strictly negatively curved manifolds. Both alternatives are excluded on every smooth type in the limiting subsequence. This proves the theorem. In particular the argument controls the possible vanishing of the lowest Weyl amplitude instead of hiding it in a nondegeneracy hypothesis.

Under the original lower-amplitude hypothesis, the spectral assumptions themselves also force the endpoint volume if the splitting tends to zero; endpoint volume is not a premise of Theorem 1. Einstein Chern–Weil gives

$$\frac{2\pi^2 T}{V} = \frac{1}{3} + \fint |W^+|^2 = 1 + \frac{8}{3} \fint (t^2 - \tfrac{1}{4}) + \frac{1}{2}\mathscr{S}. \tag{10}$$

By (6) and (8), the last two terms are bounded in absolute value by $C\sqrt{\mathscr{S}} + \mathscr{S}/2$. When $T = A$, (5) also gives $V \geqslant 2\pi^2 T$. Consequently, for sufficiently small splitting,

$$0 \leqslant \frac{V}{2\pi^2 T} - 1 \leqslant C_{m,\theta}\sqrt{\mathscr{S}}. \tag{11}$$

Thus the hypotheses cover the full volume window before showing that a vanishing-splitting sequence must approach its lower end. They do not exclude candidates whose splitting is bounded away from zero.

5. Opposite-Weyl degeneration without the aspherical input

There is a more precise necessary degeneration picture in the general fixed-topology, bounded-diameter case. This is no longer a remaining possibility for the aspherical self-gluings: the compact-core obstruction excludes their bubbles directly, without proving compatibility of complex fillings. Suppose g_j are exact Einstein, satisfy (1), $\mathscr{S}(g_j) \to 0$, and have bounded diameter, but no uniform W^- bound.

The volume lower bound is still (5). The four-dimensional Einstein Chern–Gauss–Bonnet identity gives

$$8\pi^2\chi(M) = \int_M \left(|W^+|^2 + |W^-|^2 + \frac{2}{3} \right), \qquad \mathrm{Vol}(g_j) \leqslant 12\pi^2\chi(M). \tag{9}$$

Thus the full curvature energy is bounded and relative Bishop comparison gives noncollapse. Classical Einstein orbifold compactness supplies an Einstein orbifold limit with smooth convergence off finitely many points, together with the usual ALE rescaling analysis at concentrations.

Because (9) bounds total volume, the unnormalized splitting integral also tends to zero. On the regular part of the limit, (8) gives

$$t_\infty = \tfrac{1}{2}, \qquad \nabla\omega_\infty = 0.$$

The lowest eigenline extends as an orbifold eigenline since its eigenvalue stays simple and nonzero. Its orientation orbifold cover therefore carries the induced negative Kähler–Einstein orbifold metric.

At a curvature blow-up scale $\rho_j \to 0$, the rescaled metrics are $\rho_j^{-2} g_j$. Their Ricci constant tends to zero, while (7) gives

$$\|W^+_{\rho_j^{-2} g_j}\|_\infty \leqslant C_\theta \rho_j^2 \longrightarrow 0.$$

Consequently any smooth nonflat ALE bubble obtained in this process is Ricci-flat and anti-self-dual in the original orientation. Wright's classification shows that such smooth bubbles are hyperkähler or suitable cyclic quotients of Gibbons–Hawking spaces, and are Kähler.

This is **not** a proof that the original M is Kähler. A Kähler orbifold and Kähler ALE pieces do not by themselves provide compatible complex structures across the actual marked necks, a global complex smoothing, or the original smooth type. Intermediate bubble-tree objects can themselves be orbifolds. No assertion that all bubbles are simply connected, nor that a global complex smoothing is unobstructed, is used. Thus, in the absence of an additional topological argument, opposite-chirality concentration remains a genuine mechanism; it is not silently removed by the word "resolution". On the actual aspherical self-gluings, the later compact-core argument supplies precisely such an additional argument and rules it out.

The lower amplitude hypothesis is used in this specific KE-orbifold description. Without that hypothesis, an orbifold limit in a general family can instead have $W^+ \equiv 0$. One may not transfer $\tau(M) = 0$ directly to its orbifold signature and repeat the smooth-limit hyperbolicity argument: bubble and boundary correction terms intervene. The smooth gap of Theorem 3 is unaffected.

6. Why the universal small-splitting vanishing theorem remains open here

The signed identity in the preceding work equates positive connection energy to positive splitting energy. It does not force either to vanish. At the negative KE curvature model the splitting reaction has coefficient -6, so there is no immediate positive-mass absorption proof.

Near a fixed negative KE metric, Dai–Wang–Wei, Theorem 3.6, prove that nearby Einstein metrics remain KE when all infinitesimal complex deformations are integrable. This is a genuine sufficient geometric condition, not an invertibility assumption. It does not give a uniform statement over degenerating moduli or over obstructed complex structures. Nagy's 2026 second- and third-order deformation results are finite-order results; they cannot be substituted for all-order local integrability.

The present theorem avoids that unresolved local issue by using actual non-virtual-Kähler topology. It excludes smooth compact limits on those objects regardless of the deformation theory of any hypothetical KE limit. Diameter escape is not removed by either this note or its aspherical strengthening. The marked ALE compatibility problem is not solved here in general; the later aspherical result bypasses it for the actual self-gluings by proving that an ALE core cannot embed at all.

7. Primary sources and provenance

- A. Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*, Compositio Math. 49 (1983), Proposition 5 and Theorem 2: https://www.numdam.org/article/CM_1983___49_3_405_0.pdf.
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- X. Dai, X. Wang, G. Wei, *On the Variational Stability of Kähler-Einstein Metrics*, Theorem 3.6 in the inspected author manuscript: https://web.math.ucsb.edu/~dai/paper/dww3.pdf.
- P.-A. Nagy, second-order and third-order negative KE deformation results: https://arxiv.org/abs/2603.09028 and https://arxiv.org/abs/2606.04501.

The application uses the complete hypotheses of the inherited shear topology and actual monopole results. The new content of this note is their composition with the spectral structure estimate to obtain the global escape alternative, including the explicit zero-budget-ratio application. It introduces neither a new compactness theorem nor a general Einstein nonexistence result.